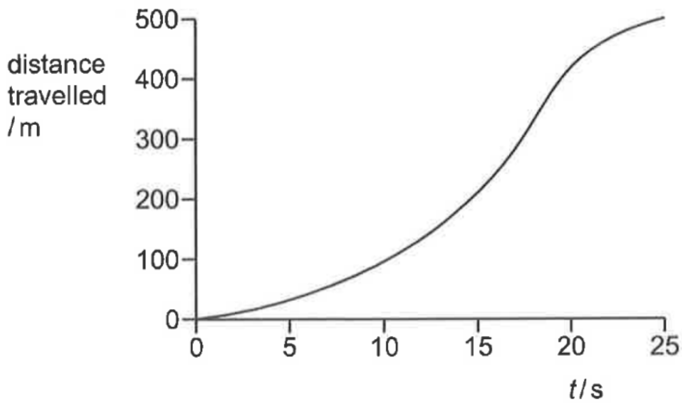
**5** A car, starting from rest at time $t = 0$, travels along a road. The distance travelled from the starting point is measured over the next 25 seconds.




Which best describes the motion of the car?

**A** The maximum speed during the first 20 seconds is 10 m $s^{-1}$.

**B** At some instant during the first 20 seconds the speed is exactly 20 m $s^{-1}$.

**C** The average speed for the first 200 m of the journey is 20 m $s^{-1}$.

**D** The average speed between 20 and 25 seconds is greater than that between 15 and 20 seconds.

**Ans: B**
Option A is wrong as it covered 400 m in 20 s. The average speed is already at 20 m $s^{-1}$. Thus the max speed must be higher.
Option C is wrong. It took 15 s to cover 200 m. Thus, the average speed is 13.3 m $s^{-1}$.
Option D is wrong. From 20 to 25 s, 100 m was covered. From 15 to 20 s, 200 m was covered, thus the average speed from 15 to 20 s must be higher.
Option B is correct. The average speed over 20 s is 20 m $s^{-1}$. At 0 s, it started at a low speed, at 20 s the instantaneous speed is higher than 20 m $s^{-1}$. As speed is a continuous variable (no abrupt change, jumping from one value to another without going through values inbetween), thus at some instant during the 20 s, the speed must have been 20 m $s^{-1}$.